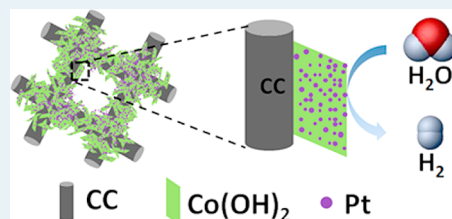


Ultrafine Pt Nanoparticle-Decorated $\text{Co}(\text{OH})_2$ Nanosheet Arrays with Enhanced Catalytic Activity toward Hydrogen EvolutionZhikai Xing,[†] Ce Han,[†] Dewen Wang,^{†,‡} Qun Li,^{†,‡} and Xiurong Yang^{*,†}[†]State Key Laboratory of Electroanalytical Chemistry, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, Jilin China[‡]University of Science and Technology of China, Hefei 230026, China

Supporting Information

ABSTRACT: The combinations of Earth-abundant materials with noble metals provide an orientation for developing highly active and stable catalysts toward hydrogen production with reduced noble metal loadings. Here, we designed carbon cloth (CC)-supported Earth-abundant $\text{Co}(\text{OH})_2$ nanosheets array ($\text{Co}(\text{OH})_2/\text{CC}$) as an ideal three-dimensional (3D) substrate for Pt electrodeposition ($\text{Pt}-\text{Co}(\text{OH})_2/\text{CC}$, Pt in $\text{Pt}-\text{Co}(\text{OH})_2$: 5.7 wt %) to achieve top performance of a hydrogen evolution reaction (HER) under alkaline and neutral conditions. The $\text{Pt}-\text{Co}(\text{OH})_2/\text{CC}$ catalyst exhibits a near-zero onset overpotential and a Tafel slope of 70 mV dec^{-1} , and it requires an overpotential of 32, 54, and 122 mV to deliver the geometrical current density of 10, 20, and 100 mA cm^{-2} , respectively, with catalytic activities exceeding to those of the commercial Pt/C decorated CC (Pt/C/CC). Furthermore, the HER activity of $\text{Co}(\text{OH})_2$ decorated with several transition metals (Ni, Co, and Fe) was demonstrated in experiments, further validating the high HER activity of the $\text{Pt}-\text{Co}(\text{OH})_2/\text{CC}$ catalyst. In addition, this catalyst also offers enhanced catalytic performance and durability under neutral conditions. Impressively, based on the normalized HER current densities by electrochemical surface area, the HER activity of the $\text{Pt}-\text{Co}(\text{OH})_2/\text{CC}$ catalyst is 4.8 and 2.6 times greater than that of the commercial Pt/C/CC in alkaline and neutral solution, respectively. The unprecedented catalytic performances of $\text{Pt}-\text{Co}(\text{OH})_2/\text{CC}$ catalyst are attributed to the synergistic catalytic effects originating from the nanointerfaces between $\text{Co}(\text{OH})_2$ and Pt.



KEYWORDS: $\text{Co}(\text{OH})_2$, Pt nanoparticles, reduced Pt loading, electrocatalyst, hydrogen evolution reaction

With increasing energy costs and concerning global warming and climate change, hydrogen as a clean and renewable energy resource has gained grand attention for tackling the current energy crisis and environmental issues.^{1,2} Alkaline water electrolysis, because of its suitable manufacture safety, stable output, and high product purity, is considered as one of the most sustainable approaches for producing hydrogen.^{3–5} However, because of the inhibition of sluggish kinetics of hydrogen evolution reaction (HER) in alkaline media, the electrocatalysts always exhibit high overpotential and large energy consumption, further restricting their practical application in alkaline electrolysis.^{6,7} Platinum (Pt) as the state-of-art electrocatalyst for alkaline HER, but its activity is 2 orders of magnitude lower than that for acidic HER.^{6–8} In addition, despite the great success that has been achieved based on Earth-abundant electrocatalysts that include transition-metal nitrides, sulfides, and phosphides for alkaline HER,² Pt-based electrocatalysts are still the preferred HER catalysts with unparalleled activity. Therefore, new concepts that facilitate the development of catalysts with several unique features, including high Pt utilization, enhanced activity, and improved stability are still challenging.

The combinations of the individual advantages of Pt and Earth-abundant materials have emerged as an effective strategy to enhance the HER activity of Pt in an alkaline environ-

ment.^{7,9–14} Recently, the Markovic group electrochemically deposited $\text{Ni}(\text{OH})_2$ nanoclusters on a Pt electrode surface for catalyzing HER.⁷ In their bifunctional materials, the edges of $\text{Ni}(\text{OH})_2$ clusters promote the process of water dissociation and the generation of hydrogen intermediates that are subsequently adsorbed on the nearby Pt surface and finally recombines to form molecular hydrogen.⁷ Following this strategy, Chen and co-workers further modified a Pt_3Ni nanoframe via the electrodeposition of $\text{Ni}(\text{OH})_2$ clusters for increasing the HER activity in alkaline electrolytes.⁹ Based on these two works, Yin and co-workers synthesized ultrathin Pt nanowires (NWs) on single-layer (SL) $\text{Ni}(\text{OH})_2$ nanosheets hybrid material (Pt content in the hybrid is 38.0 wt %) for constructing high-performance HER catalysts in basic solution.¹⁰ Recently, Zeng and co-workers further demonstrated that Ni (hydroxy)oxide/Pt(111) film had enhanced HER activity through a bifunctional mechanism, using a combination of density functional theory and in situ characterization.¹³ However, all these catalysts involved the massive consumption of the noble metal platinum (Pt as substrates;^{7,9,11–13} Pt in Pt NWs/SL- $\text{Ni}(\text{OH})_2$ is 38 wt %;¹⁰ Pt in $\text{Pt}@2\text{D}-\text{Ni}(\text{OH})_2$ is 43

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wt %¹⁴), which impedes its commercialization, thus offering us a new strategy for the development of novel Pt-based catalysts with distinguished activity and durability, while reducing the overall Pt loading. Moreover, nanoarray catalysts have their intrinsic advantages, including good electrical conductivity and high structural stability, large surface area for efficient utilization of active species, and reduced catalyst loading with enhanced electrochemical performance.^{15–19} Therefore, in this work, we report on the development of Pt–Co(OH)₂ nanosheet arrays supported on carbon cloth (CC) (Pt–Co(OH)₂/CC, Pt in Pt–Co(OH)₂ is 5.7 wt %) for exploiting the synergistic effects of Co(OH)₂ and Pt for the electrochemical HER. The Co(OH)₂/CC nanosheets array with low-cost and high surface area was not only employed as an ideal 3D substrate for Pt electrodeposition with high utilization rate of Pt, but also served as a new form of synergistic catalyst for tremendously enhanced hydrogen generation. As a proof-of-concept, the obtained Pt–Co(OH)₂/CC catalyst exhibits unprecedented catalytic activity and durability toward HER under basic and neutral conditions.

The Pt–Co(OH)₂/CC electrocatalyst was fabricated via the hydrothermal synthesis of Co(OH)₂/CC, followed by the electrodeposition of Pt, as illustrated in Figure 1. CC was

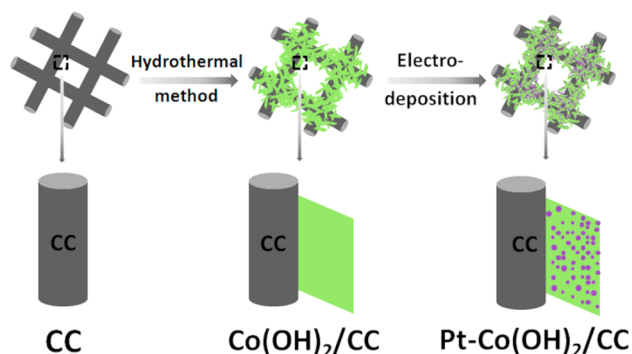


Figure 1. Fabrication process of a Pt–Co(OH)₂/CC electrocatalyst.

applied as a three-dimensional (3D) support for integrating HER catalysts; a representative scanning electron microscopy (SEM) image of CC is shown in Figure S1a in the Supporting Information. Co(OH)₂ nanosheets were densely grown on the surface of CC to form Co(OH)₂/CC nanosheet arrays (see Figures S1b–S1d in the Supporting Information), and such 3D Co(OH)₂/CC substrates exhibit high porosity with large Brunauer–Emmett–Teller (BET) surface areas (up to 71.6 m² g^{−1}) (see Figure S2 in the Supporting Information). Then, Pt was electrodeposited on the surface of the Co(OH)₂/CC to form Pt–Co(OH)₂/CC nanosheet arrays. The X-ray diffraction (XRD) pattern of Pt–Co(OH)₂/CC shows that no Pt-containing crystal phases can be observed (see Figure 2a),²⁰ because the Pt content in Pt–Co(OH)₂ nanosheets array is ~5.7 wt % (<2 wt % when taking the CC into account), which is quantitatively measured with an inductively coupled plasma mass spectrometry (ICP-MS) technique. According to the SEM images (see Figures 2b and 2c) and transmission electron microscopy (TEM) image (Figure 2d), no large Pt particles were observed, which was consistent with scanning tunnelling electron microscopy (STEM) images and energy-dispersive X-ray (EDX) mapping images showing the homogeneous distribution of Co, O, and Pt over the nanosheet (Figure S3 in the Supporting Information). After electrodeposition of Pt,

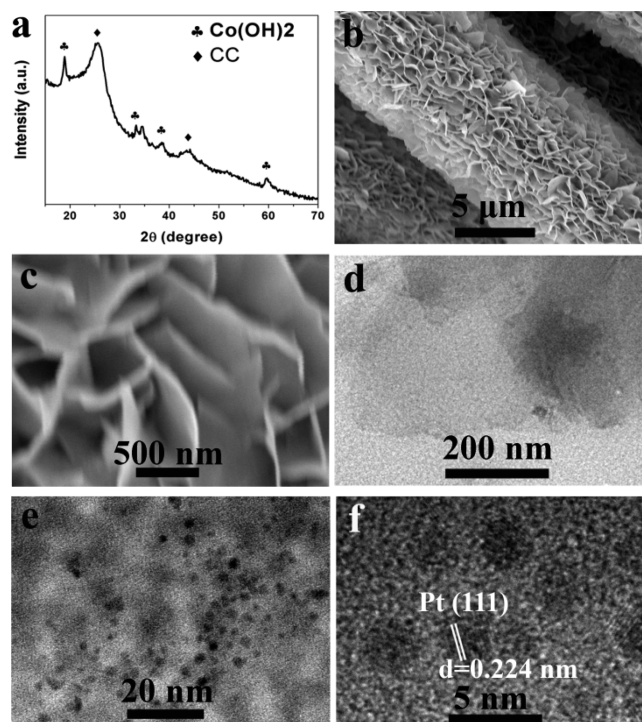


Figure 2. (a) XRD pattern of Pt–Co(OH)₂/CC. (b, c) SEM images of Pt–Co(OH)₂/CC. (d, e) TEM and (f) HRTEM images of Pt–Co(OH)₂.

smaller nanoparticles are observed on a Co(OH)₂ nanosheet with diameters ranging from 2.0 nm to 3.5 nm (see Figure 2e). The high-resolution TEM image shows that the lattice spacing of ~0.224 nm agreed well with the (111) plane of Pt (see Figure 2f).^{21–23} Figure S4 in the Supporting Information shows the X-ray photoelectron spectroscopy (XPS) spectrum in the Pt 4f region for Pt–Co(OH)₂/CC. The binding energies of Pt 4f_{7/2} and Pt 4f_{5/2} of Pt–Co(OH)₂ were 71.0 and 74.4 eV, respectively, which are similar to the binding energies of previously reported Pt-based catalysts,^{21,23} suggesting that all the Pt nanoparticles coated on the Co(OH)₂ nanosheet arrays are mainly metallic Pt. The above results demonstrate that smaller Pt nanoparticles were successfully decorated on the surface of Co(OH)₂/CC nanosheet arrays via electrodeposition techniques.

The electrocatalytic activities of the Pt–Co(OH)₂/CC (Pt–Co(OH)₂ loading: 6.9 mg cm^{−2}) toward HER normalized by geometric surface area (GSA, 1 cm × 1 cm) along with Pt-decorated CC (Pt/CC), Co(OH)₂/CC, blank CC, and commercial Pt/C (20 wt % Pt/XC-72) loaded CC (Pt/C/CC) catalysts were performed in N₂-saturated 1.0 M KOH solution to verify the synergistic effects (see Figure 3). The Nyquist plots of Pt–Co(OH)₂/CC and Co(OH)₂/CC shown in Figure 3a, as well as Figure S5 in the Supporting Information indicate that the ionic and ohmic resistances of Pt–Co(OH)₂/CC are much smaller than those of Co(OH)₂/CC, implying that the electrodeposition of Pt can obviously enhance the electronic conductivity of Co(OH)₂/CC. In the linear sweep voltammetry (LSV) curves on the reversible hydrogen electrode (RHE) scale, a near-zero onset overpotential for the Pt–Co(OH)₂/CC was observed, which was ~210 mV lower than that of Co(OH)₂/CC (Figure 3b). A sharp increase in current density was observed when compared to those of Co(OH)₂/CC and Pt/CC, corresponding to more rapid

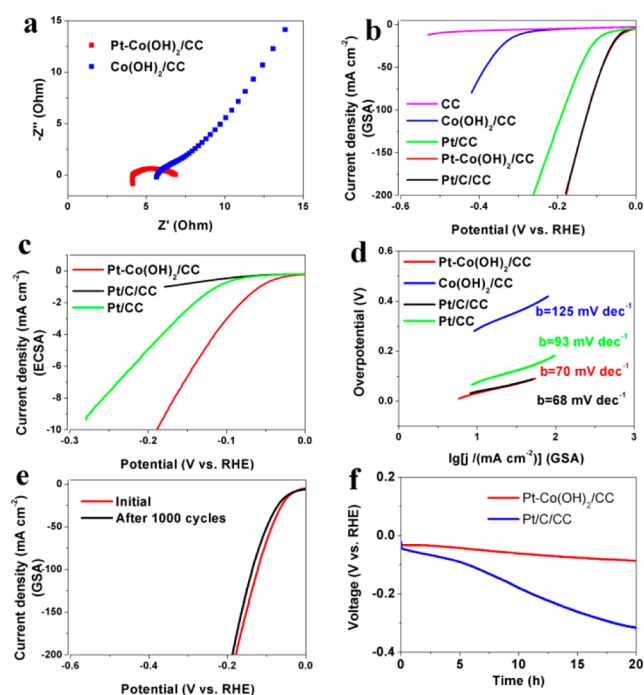


Figure 3. (a) Nyquist plots of electrochemical impedance spectra of $\text{Co(OH)}_2/\text{CC}$ and $\text{Pt-Co(OH)}_2/\text{CC}$ at $\eta = 100$ mV in 1.0 M KOH. (b) Polarization curves of CC, $\text{Co(OH)}_2/\text{CC}$, Pt/CC, $\text{Pt-Co(OH)}_2/\text{CC}$, and Pt/C/CC with current densities normalized to GSA. (c) Polarization curves of Pt/CC, $\text{Pt-Co(OH)}_2/\text{CC}$, and Pt/C/CC with current densities normalized to ECSA. (d) Tafel plots of $\text{Co(OH)}_2/\text{CC}$, Pt/CC, $\text{Pt-Co(OH)}_2/\text{CC}$, and Pt/C/CC with current densities normalized to GSA. (e) Stability of $\text{Pt-Co(OH)}_2/\text{CC}$ electrode with an initial polarization curve and after 1000 cycles at a scan rate of 100 mV s^{-1} between +0.05 and −0.25 V vs RHE in 1.0 M KOH (the current densities normalized to GSA). (f) Chronopotentiometric measurements of long-term stability of $\text{Pt-Co(OH)}_2/\text{CC}$ and Pt/C/CC at 10 mA cm^{-2} , which normalized to GSA.

hydrogen evolution. The $\text{Pt-Co(OH)}_2/\text{CC}$ catalyst approaches current densities of 10, 20, and 100 mA cm^{-2} at overpotentials of 32, 54, and 122 mV, respectively, and these values are smaller than those for Pt/C/CC (38, 56, and 123 mV) (see Figure 3b, as well as Figure S6 in the Supporting Information). In contrast, Pt/CC exhibits inferior HER activity, because overpotentials of 75, 104, and 185 mV are needed to deliver the same current densities, although its onset overpotential is similar to those of $\text{Pt-Co(OH)}_2/\text{CC}$ and Pt/C/CC (see Figure S6). The undoubtedly remarkable HER performances of $\text{Pt-Co(OH)}_2/\text{CC}$ electrocatalyst were the best among all reported values for transition-metal- and noble-metal-based HER catalysts in basic media with current densities normalized to GSA of electrodes (see Table S1 in the Supporting Information).

The LSV curves of $\text{Pt-Co(OH)}_2/\text{CC}$ under different Pt electrodeposition times are shown in Figure S7 in the Supporting Information, and its catalytic activity first increased and then decreased as the deposition time increased from 5 min to 20 min, which is consistent with the previous report.²⁴ When the electrodeposition time was 15 min, the $\text{Pt-Co(OH)}_2/\text{CC}$ catalyst exhibits the highest electrocatalytic activity. To understand the original differences in HER performance of Pt/CC, $\text{Pt-Co(OH)}_2/\text{CC}$, and Pt/C/CC, we measured the capacitances of the double-layer at the solid/liquid interface of these electrodes by collecting cyclic voltammograms (CVs) in

1.0 M KOH (Figure S8 in the Supporting Information) to estimate their electrochemical surface area (ECSA).^{18,25} Figure 3c shows the LSV curves of these three catalysts with current densities normalized to ECSA. The HER current density of $\text{Pt-Co(OH)}_2/\text{CC}$ at an overpotential of 0.07 V is 1.518 mA cm^{-2} , which is much higher than that of Pt/CC (0.361 mA cm^{-2}) and commercial Pt/C/CC (0.318 mA cm^{-2}), respectively. The performance of the $\text{Pt-Co(OH)}_2/\text{CC}$ catalyst is comparable to that observed for the currently reported electrocatalysts (the current densities normalized to ECSA) in alkaline media (see Table S2 in the Supporting Information). Note that the current density of $\text{Pt-Co(OH)}_2/\text{CC}$ is 4.2 and 4.8 times higher than that of Pt/CC and commercial Pt/C/CC counterpart, respectively. Figure 3d shows the corresponding Tafel plots for further analyzing their HER kinetics. The value of $\text{Pt-Co(OH)}_2/\text{CC}$ (70 mV dec^{-1}) is smaller than that of Pt/CC (93 mV dec^{-1}) and $\text{Co(OH)}_2/\text{CC}$ (125 mV dec^{-1}); however, it is slightly larger than that of commercial Pt/C/CC (68 mV dec^{-1}). In alkaline media, the Volmer step ($\text{H}_2\text{O} + \text{e}^- \rightarrow \text{H}_{\text{ad}} + \text{OH}^-$) is the rate-limiting step for Pt/C as HER catalysts.^{26,27} Therefore, for the $\text{Pt-Co(OH)}_2/\text{CC}$ catalyst, the Co(OH)_2 nanosheet array provides active sites for water dissociation and therefore accelerates the Volmer step under basic conditions, leading to a markedly improved HER activity.

The stability of the $\text{Pt-Co(OH)}_2/\text{CC}$ electrocatalyst during repeated cycling in alkaline electrolyte was further evaluated. The $\text{Pt-Co(OH)}_2/\text{CC}$ catalyst exhibits no obvious loss of activity after 1000 CV sweeps between +50 mV to −250 mV vs RHE at a scan rate of 100 mV s^{-1} . The overpotential changed to 136 mV to produce a current density of 100 mA cm^{-2} after 1000 cycles, whereas the corresponding overpotential initially required is 122 mV, suggesting its high stability under alkaline conditions (see Figure 3e). The morphology and structure of Pt-Co(OH)_2 nanosheet arrays can be well-retained after cycling for 1000 cycles (see Figure S9 in the Supporting Information). Figure S10 in the Supporting Information shows the XPS spectrum of the Pt 4f region for the $\text{Pt-Co(OH)}_2/\text{CC}$ electrode after 1000 cycles. The positions for the Pt 4f_{7/2} and Pt 4f_{5/2} peaks of metallic Pt exhibit almost no shift after cycling. Combined with the Pt content in Pt-Co(OH)_2 nanosheet arrays (5.5 wt %) after cycling, which measured by ICP-MS technique, these results suggest a strong interaction between the Pt and $\text{Co(OH)}_2/\text{CC}$ substrate, indicating good stability of $\text{Pt-Co(OH)}_2/\text{CC}$ catalyst. Chronopotentiometric (v - t) responses of $\text{Pt-Co(OH)}_2/\text{CC}$ and Pt/C/CC catalysts at a constant cathodic current density of 10 mA cm^{-2} were tested for 20 h (Figure 3f). For the benchmark Pt/C/CC catalyst, the potential required to deliver a current density of 10 mA cm^{-2} decreases rapidly in the chronopotentiometric curve, in agreement with our previous report.²⁸ Impressively, under the same conditions, the corresponding potential for $\text{Pt-Co(OH)}_2/\text{CC}$ catalyst shows considerably slower decay than that for commercial Pt/C/CC.¹⁰ These results confirm that Co(OH)_2 nanosheet arrays could effectively improve the long-term stability of Pt in basic media, being attributed to the superior durability of $\text{Pt-Co(OH)}_2/\text{CC}$.

Considering the anomalous enhancement of catalytic activity over $\text{Co(OH)}_2/\text{CC}$ via Pt electrodeposition, the HER activity of $\text{Co(OH)}_2/\text{CC}$ decorated with other transition metals (Ni, Co, and Fe) were also investigated. Therefore, Ni-decorated $\text{Co(OH)}_2/\text{CC}$ (Ni- $\text{Co(OH)}_2/\text{CC}$, Ni in Ni- Co(OH)_2 : 5.6 wt %, by ICP-MS), Fe- $\text{Co(OH)}_2/\text{CC}$ (Fe in Fe- Co(OH)_2 : 5.8 wt %, by ICP-MS), and Co- $\text{Co(OH)}_2/\text{CC}$ catalysts were

freshly fabricated following the similar preparation procedure as Pt-Co(OH)₂/CC. As shown in Figure 4, all these three

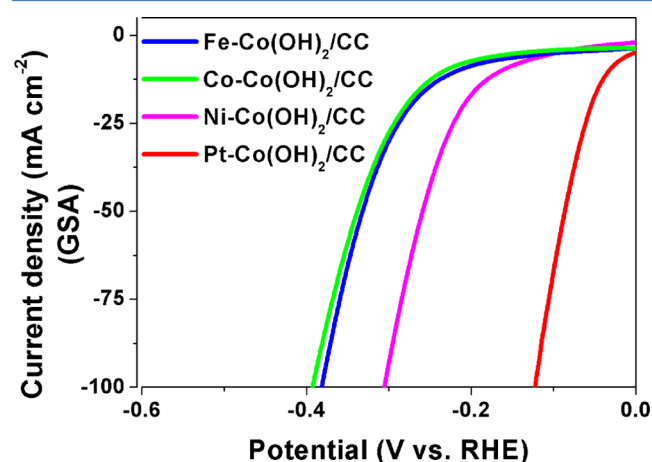


Figure 4. Polarization curves for Fe-Co(OH)₂/CC, Co-Co(OH)₂/CC, Ni-Co(OH)₂/CC, and Pt-Co(OH)₂/CC with current densities normalized to GSA in 1.0 M KOH.

catalysts exhibit higher HER activity than that of Co(OH)₂/CC (Figure 3b). Among them, Ni-Co(OH)₂/CC catalyst gives a current density of 10 mA cm⁻² at an overpotential of 163 mV, thereby largely reducing the corresponding overpotential of ~125 mV, compared with Co(OH)₂/CC. Consequently, under our experimental conditions, the trend of HER activity of these decorated catalysts is Pt-Co(OH)₂/CC > Ni-Co(OH)₂/CC > Fe-Co(OH)₂/CC > Co-Co(OH)₂/CC, which is in good agreement with their results of electrical impedance spectra (see Figure 3a, as well as Figure S11 in the Supporting Information), further corroborate that the Pt-Co(OH)₂/CC catalyst possesses the best catalytic activity and durability toward HER.

In addition to functioning as a highly efficient HER electrocatalyst in alkaline media, the Pt-Co(OH)₂/CC catalyst also served as highly active HER electrocatalysts under neutral conditions (1.0 M phosphate buffer solution, PBS). For comparison, the commercial Pt/C/CC benchmark catalyst exhibits a high HER activity with an overpotential of 42 mV at a current density of 10 mA cm⁻² (see Figure 5a). As expected, the Pt-Co(OH)₂/CC catalyst produces the current densities of 10 and 100 mA cm⁻² at overpotentials of 84 and 253 mV, respectively, which is much lower than those for Co(OH)₂/CC (328 mV) and Pt/CC (122 mV) at the current density of 10 mA cm⁻² (Figure 5a). These overpotentials are the lowest among the reported high-performance HER catalysts in neutral media (the current densities normalized to GSA), except for that of the commercial Pt/C/CC (see Table S1). As shown in Figure 5b, at an overpotential of 0.07 V, the current densities normalized to ECSA for the commercial Pt/C/CC, Pt/CC, and Pt-Co(OH)₂/CC are 0.137, 0.188, and 0.335 mA cm⁻², respectively. Note that the HER activity of Pt-Co(OH)₂/CC is 2.6 times higher than that of commercial Pt/C/CC. The Pt-Co(OH)₂/CC catalyst also possesses excellent HER stability; its catalytic activity did not change after 1000 CV sweeps (see Figure 5c).

In summary, we have developed a Pt-Co(OH)₂/CC electrocatalyst, whereby ultrafine Pt nanoparticles were intimately interfaced with Earth-abundant Co(OH)₂ nanosheets with a low noble-metal content of 5.7 wt %, for highly

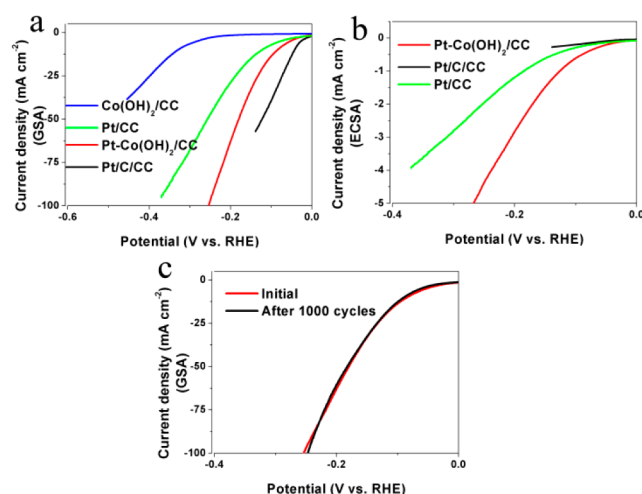


Figure 5. (a) Polarization curves of Co(OH)₂/CC, Pt/CC, Pt-Co(OH)₂/CC, and Pt/C/CC with current densities normalized to GSA in 1.0 M PBS. (b) Polarization curves of Pt/CC, Pt-Co(OH)₂/CC, and Pt/C/CC with current densities normalized to ECSA in 1.0 M PBS. (c) Stability of Pt-Co(OH)₂/CC electrode with an initial polarization curve and after 1000 cycles at a scan rate of 100 mV s⁻¹ between +0.05 and -0.25 V vs RHE in 1.0 M PBS (the current densities normalized to GSA).

efficient hydrogen evolution. Our experimental results demonstrate that the Pt-Co(OH)₂/CC catalyst exhibits an almost-zero onset overpotential and a top-level electrocatalytic activity for HER, which is 4.8 and 2.6 times higher than that of the commercial Pt/C in alkaline and neutral electrolytes, respectively. More importantly, regarding the similarity of the qualitative trends for Co (hydroxy)oxide and Ni (hydroxy)-oxide films on Pt (111),^{13,14} the unprecedented catalytic performances of the Pt-Co(OH)₂/CC electrocatalyst are attributed primarily to the synergistic effect of the Co(OH)₂ nanosheet array and high utilized Pt.^{13,14,28–30} To our knowledge, this work provides the first systematic study of how metal hydroxide nanosheet arrays can be applied as ideal 3D substrates for growing noble metal to enhance their electrochemical HER activity, and, therefore, it offers a new strategy for designing efficient and multifunctional electrocatalysts for HER, oxygen evolution and reduction reactions, and beyond.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.7b01994.

SEM images and XRD pattern (Figure S1); BJH pore-size distribution curve and nitrogen adsorption/desorption isotherm plot (Figure S2); STEM image and EDX elemental mapping images (Figure S3); XPS spectra (Figures S4 and S10); Nyquist plots (Figures S5 and S11); polarization curves (Figures S6 and S7); comparison of HER performance of catalysts under basic and neutral conditions (Table S1); cyclic voltammograms and the corresponding capacitive currents (Figure S8); comparison of HER performance of high-performance HER catalysts under alkaline conditions (Table S2) (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Dresselhaus, M. S.; Thomas, I. L. *Nature* **2001**, *414*, 332–337.
- (2) Zou, X.; Zhang, Y. *Chem. Soc. Rev.* **2015**, *44*, 5148–5180.
- (3) Kong, D.; Wang, H.; Lu, Z.; Cui, Y. *J. Am. Chem. Soc.* **2014**, *136*, 4897–4900.
- (4) Wang, D.; Gong, M.; Chou, H.; Pan, C.; Chen, H.; Wu, Y.; Lin, M.; Guan, M.; Yang, J.; Chen, C.; Wang, Y.; Hwang, B.; Chen, C.; Dai, H. *J. Am. Chem. Soc.* **2015**, *137*, 1587–1592.
- (5) Yang, Y.; Forster, M.; Ling, Y.; Wang, G.; Zhai, T.; Tong, Y.; Cowan, A. J.; Li, Y. *Angew. Chem.* **2016**, *128*, 3464–3468.
- (6) Durst, J. J.; Siebel, A.; Simon, C.; Hasche, F.; Herranz, J.; Gasteiger, H. A. *Energy Environ. Sci.* **2014**, *7*, 2255–2260.
- (7) Subbaraman, R.; Tripkovic, D.; Strmcnik, D.; Chang, K.; Uchimura, M.; Paulikas, A. P.; Stamenkovic, V.; Markovic, N. M. *Science* **2011**, *334*, 1256–1260.
- (8) Sheng, W.; Myint, M.; Chen, J.; Yan, Y. *Energy Environ. Sci.* **2013**, *6*, 1509–1512.
- (9) Chen, C.; Kang, Y.; Huo, Z.; Zhu, Z.; Huang, W.; Xin, H.; Snyder, J. D.; Li, D.; Herron, J. A.; Mavrikakis, M.; Chi, M.; More, K. L.; Li, Y.; Markovic, N. M.; Somorjai, G. A.; Yang, P.; Stamenkovic, V. *Science* **2014**, *343*, 1339–1343.
- (10) Yin, H.; Zhao, S.; Zhao, K.; Muqsit, A.; Tang, H.; Chang, L.; Zhao, H.; Gao, Y.; Tang, Z. *Nat. Commun.* **2015**, *6*, 6430.
- (11) Subbaraman, R.; Tripkovic, D.; Chang, K.; Strmcnik, D.; Paulikas, A. P.; Hirunsit, P.; Chan, M.; Greeley, J.; Stamenkovic, V.; Markovic, N. M. *Nat. Mater.* **2012**, *11*, 550–557.
- (12) Stamenkovic, V. R.; Strmcnik, D.; Lopes, P. P.; Markovic, N. M. *Nat. Mater.* **2016**, *16*, 57–69.
- (13) Zeng, Z.; Chang, K.; Kubal, J.; Markovic, N. M.; Greeley, J. *Nat. Energy* **2017**, *2*, 17070.
- (14) Wang, L.; Zhu, Y.; Zeng, Z.; Lin, C.; Giroux, M.; Jiang, L.; Han, Y.; Greeley, J.; Wang, C.; Jin, J. *Nano Energy* **2017**, *31*, 456–461.
- (15) Ma, T.; Dai, S.; Jaroniec, M.; Qiao, S. J. *J. Am. Chem. Soc.* **2014**, *136*, 13925–13931.
- (16) Faber, M. S.; Dziedzic, R. D.; Lukowski, M. A.; Kaiser, N. S.; Ding, Q.; Jin, S. *J. Am. Chem. Soc.* **2014**, *136*, 10053–10061.
- (17) Feng, J.; Ding, L.; Ye, S.; He, X.; Xu, H.; Tong, Y.; Li, G. *Adv. Mater.* **2015**, *27*, 7051–7057.
- (18) Tian, J.; Liu, Q.; Asiri, A. M.; Sun, X. *J. Am. Chem. Soc.* **2014**, *136*, 7587–7590.
- (19) Tong, S.; Wang, X.; Li, Q.; Han, X. *Chin. J. Anal. Chem.* **2016**, *44*, 1447–1457.
- (20) Deng, J.; Li, H.; Xiao, J.; Tu, Y.; Deng, D.; Yang, H.; Tian, H.; Li, J.; Ren, P.; Bao, X. *Energy Environ. Sci.* **2015**, *8*, 1594–1601.
- (21) Xu, H.; Ding, L.; Feng, J.; Li, G. *Chem. Sci.* **2015**, *6*, 6991–6998.
- (22) Zheng, D.; Cao, X.; Wang, X. *Angew. Chem., Int. Ed.* **2016**, *55*, 11512–11516.
- (23) Huang, W.; Wang, H.; Zhou, J.; Wang, J.; Duchesne, P. N.; Muir, D.; Zhang, P.; Han, N.; Zhao, F.; Zeng, M.; Zhong, J.; Jin, C.; Li, Y.; Lee, S.; Dai, H. *Nat. Commun.* **2015**, *6*, 10035.
- (24) Gutic, S. J.; Dobrota, A. S.; Leetmaa, M.; Skorodumova, N. V.; Mentus, S. V.; Pasti, I. A. *Phys. Chem. Chem. Phys.* **2017**, *19*, 13281–13293.
- (25) Caban-Acevedo, M.; Stone, M. L.; Schmidt, J. R.; Thomas, J. G.; Ding, Q.; Chang, H. C.; Tsai, M.; He, J.; Jin, S. *Nat. Mater.* **2015**, *14*, 1245–1251.
- (26) Durst, J.; Siebel, A.; Simon, C.; Hasche, F.; Herranz, J.; Gasteiger, H. A. *Energy Environ. Sci.* **2014**, *7*, 2255–2260.
- (27) Lu, Q.; Hutchings, G. S.; Yu, W.; Zhou, Y.; Forest, R. V.; Tao, R.; Rosen, J.; Yonemoto, B. T.; Cao, Z.; Zheng, H.; Xiao, J. Q.; Jiao, F.; Chen, J. G. *Nat. Commun.* **2015**, *6*, 6567.
- (28) Xing, Z.; Gan, L.; Wang, J.; Yang, X. *J. Mater. Chem. A* **2017**, *5*, 7744–7748.
- (29) Zhu, H.; Zhang, J.; Yanzhang, R.; Du, M.; Wang, Q.; Gao, G.; Wu, J.; Wu, G.; Zhang, M.; Liu, B.; Yao, J.; Zhang, X. *Mater.* **2015**, *27*, 4752–4729.
- (30) Wang, Y.; Chen, L.; Yu, X.; Wang, Y.; Zheng, G. S. *Adv. Energy Mater.* **2017**, *7*, 1601390.